ECE 340: Semiconductor Devices

Lecture 9: Carrier Concentrations & Drift

Prof. Hyunseok Kim

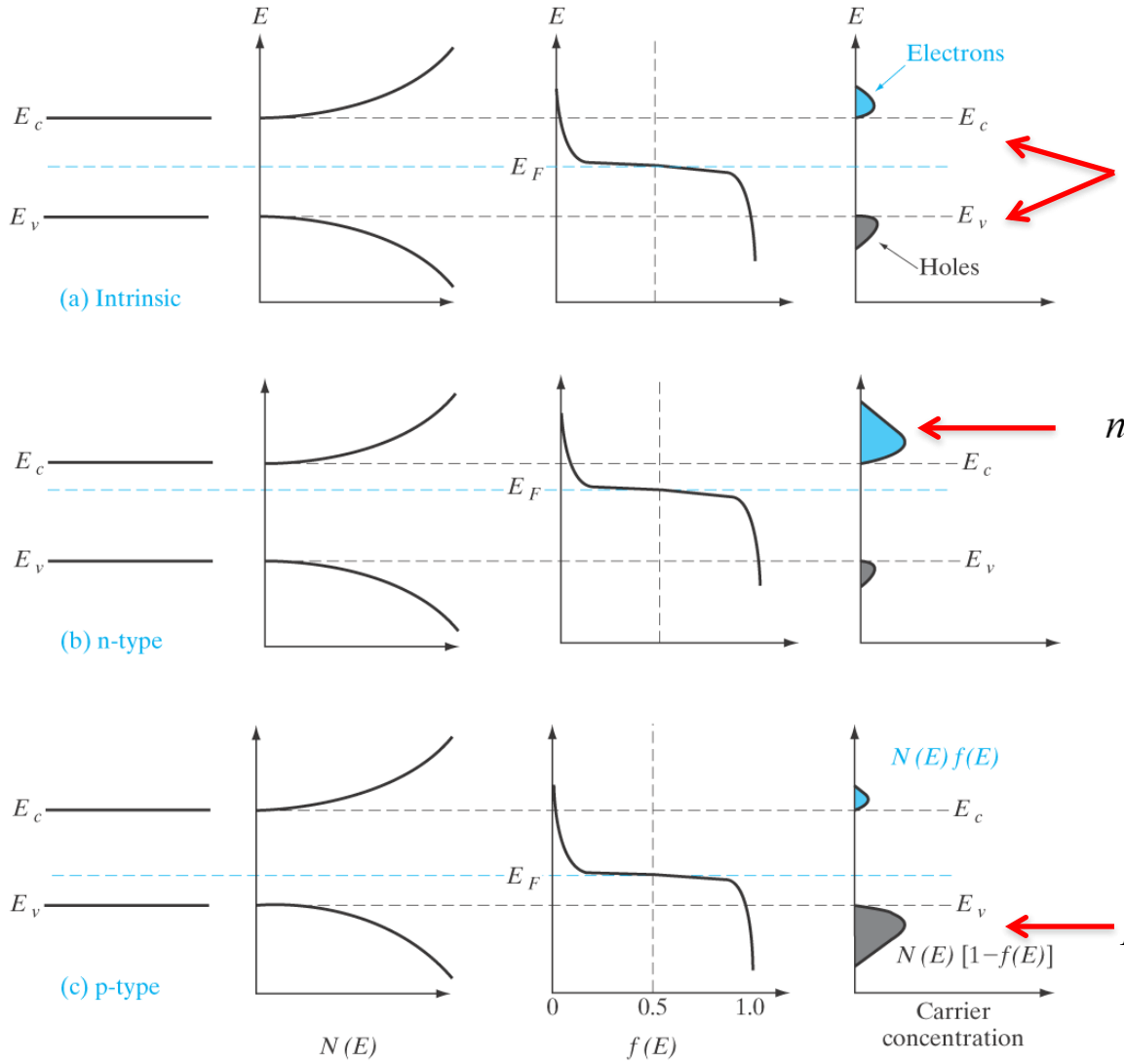
Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

E-mail: hyunseok@illinois.edu

Office Hours: Thursdays 12p-1p, 2256 HMNTL

(Recap) Carrier Concentration

The Fermi Function Multiplied by the Density of States



n_i, p_i: intrinsic concentration

Conduction band DOS

$$n_i = \int_{E_c}^{\infty} f(E) N_c(E) dE$$

$$= p_i = \int_{-\infty}^{E_V} [1 - f(E)] N_V(E) dE$$

Probability the state is empty

Valence band DOS

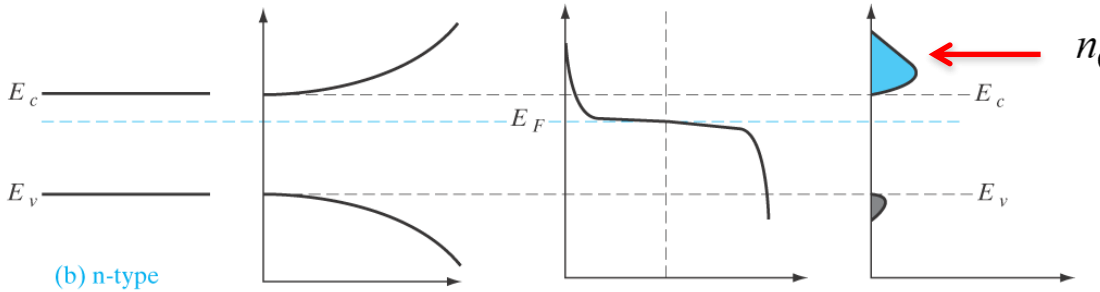
$$n_0 = \int_{E_c}^{\infty} f(E) N_c(E) dE$$

n_0, p_0: equilibrium concentration (could be intrinsic or extrinsic)

Thus, $n_0 = p_0 = n_i$ for intrinsic Si

$$p_0 = \int_{-\infty}^{E_v} [1 - f(E)] N_v(E) dE$$

(Recap) Carrier Concentration n_o at Equilibrium



$$n_0 = \int_{E_c}^{\infty} f(E) N_c(E) dE$$

$$= \int_{E_c}^{\infty} \frac{1}{1+e^{(E-E_f)/kT}} \bullet \frac{\sqrt{2}}{\pi^2} \left(\frac{m^*}{\hbar^2} \right)^{3/2} E^{1/2} dE$$

$$f(E_c) = e^{-(E_c-E_f)/kT} \qquad if \qquad e^{(E_c-E_f)/kT} >> 1$$

$$n_o = \int_{E_c}^{\infty} e^{-(E-E_f)/kT} \bullet \frac{\sqrt{2}}{\pi^2} \left(\frac{m^*}{\hbar^2} \right)^{3/2} E^{1/2} dE = 2 \left(\frac{2\pi m_n^* kT}{h^2} \right)^{3/2} e^{(E_F-E_C)/kT}$$

$$= N_C(T) e^{-(E_C-E_F)/kT}$$

$$n_o = N_c \bullet e^{-(E_c-E_f)/kT}$$

where $N_c(T) = 2 \left(\frac{2\pi \bullet m_n^* kT}{h^2} \right)^{3/2}$

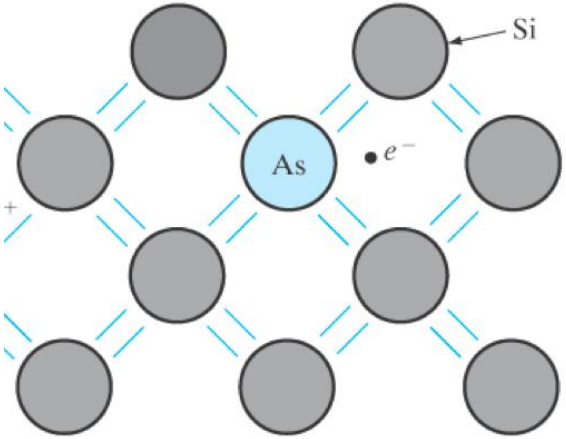
Similarly, $p_o = N_v \bullet e^{-(E_f-E_v)/kT}$

where $N_v(T) = 2 \left(\frac{2\pi \bullet m_p^* kT}{h^2} \right)^{3/2}$

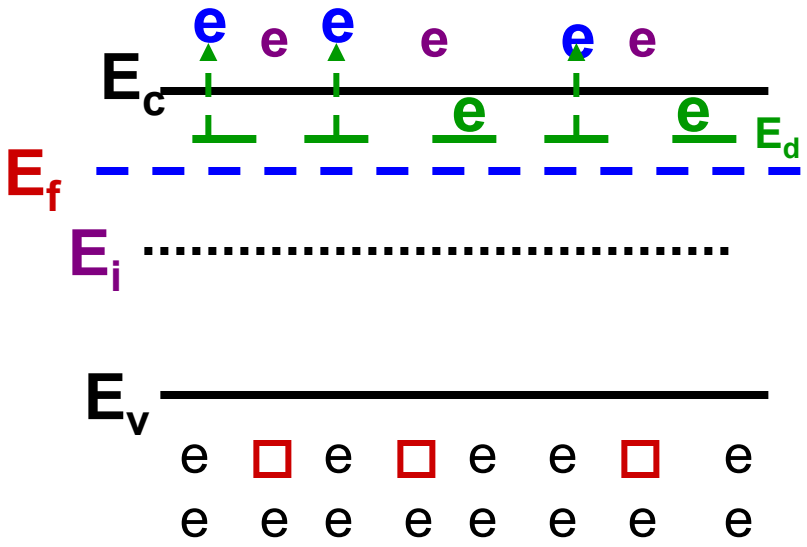
Complete solution : Appendix IV

(Recap) Electron Concentration n_o in a n-type Semiconductor at Equilibrium

Donor concentration N_D



$n_o(K)=N_D$, majority carrier



$p_o(K)=n_i^2/n_o$, minority carrier

We know $$n_o = N_c \bullet e^{-(E_c - E_f)/kT}$$ → (E_c-E_f) can be derived.

where $$N_c(T) = 2\left(\frac{2\pi \bullet m_n^* kT}{h^2}\right)^{3/2}$$

Compensation and Space Charge Neutrality

What Happens if the Crystal is Doped With Both Donors and Acceptors?

Charge Neutrality Equation :

$$p_o + N_d^+ = n_o + N_a^-$$

N_d^+ : ionized donor (positively charged)
N_a^- : ionized acceptor (negatively charged)

Example Revisited:

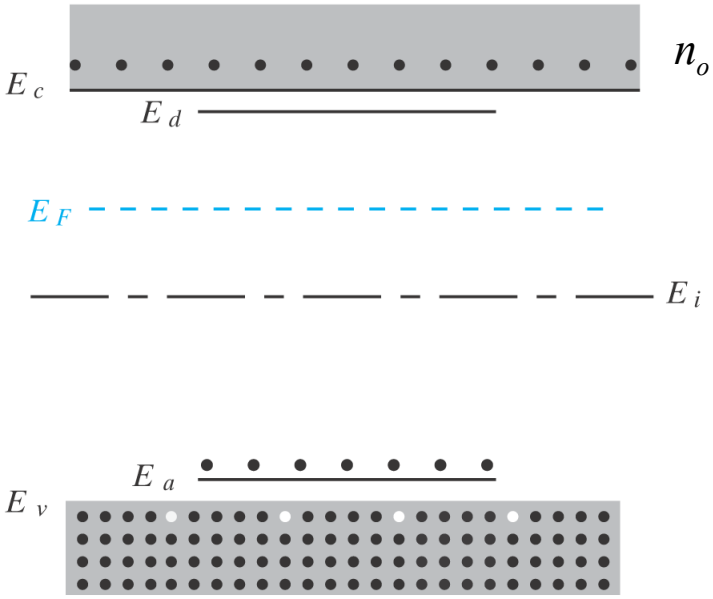
Donor-Doped Silicon, 10^{17} cm^{-3}

Since $N_d(As) >> n_i$,

$$n_o \approx N_d = 10^{17}\,cm^{-3}$$

$$p_o = \frac{n_i^2}{n_o} = \frac{\left(1.5 \times 10^{10}\,cm^{-3}\right)^2}{10^{17}\,cm^{-3}}$$

$$p_o = 2.25 \times 10^{3}\,cm^{-3} << 10^{17}\,cm^{-3}$$



$$n_o = \left(N_d^+ - N_a^-\right) + p_o \approx N_d^+ - N_a^-$$

Figure 3.19

Compensation in an n-type semiconductor ($N_d > N_a$).

From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.
ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

What Happens if the Crystal is Doped With Both Donors and Acceptors?

Charge Neutrality Equation

$$p_o + N_d^+ = n_o + N_a^-$$

Case 1: $N_D > N_A$ and $N_D - N_A >> n_i$

Given GaAs at 300K with $n_{i,GaAs} = 2\times10^6\ cm^{-3}$

$N_D = 1\times10^{17}\ cm^{-3}$

$N_A = 5\times10^{16}\ cm^{-3}$

Then:

$$n_o \approx n_o - p_o = N_D^+ - N_A^- = 1\times10^{17} - 5\times10^{16}\ cm^{-3}$$

$$n_o \approx 5\times10^{16}\ cm^{-3}$$

$$p_o = \frac{n_i^2}{n_o} \approx \frac{\left(2\times10^6\right)^2}{5\times10^{16}} = 8\times10^{-5}\ cm^{-3}$$

Case 2: $N_A > N_D$ and $N_A - N_D >> n_i$

Given Si at 300K with $n_{i,Si} = 1.5\times10^{10}\ cm^{-3}$

$N_A = 1\times10^{17}\ cm^{-3}$

$N_D = 5\times10^{16}\ cm^{-3}$

Then:

$$p_o \approx p_o - n_o = N_A^- - N_D^+ = 1\times10^{17} - 5\times10^{16}\ cm^{-3}$$

$$p_o \approx 5\times10^{16}\ cm^{-3}$$

$$n_o = \frac{n_i^2}{p_o} \approx \frac{\left(1.5\times10^{10}\right)^2}{5\times10^{16}} = 4.5\times10^{3}\ cm^{-3}$$

From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.
ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

What Happens Close to the Intrinsic Concentration?

Case 3: $N_D > N_A$ and $N_D - N_A \approx n_i$

Given Ge at 300K with $N_D = 1 \times 10^{14}\, cm^{-3}$ and $N_A = 7 \times 10^{13}\, cm^{-3}$

Then: $n_{i,Ge} = 2.5 \times 10^{13}\ cm^{-3}$

$$n_o - p_o = N_D^+ - N_A^- = 1 \times 10^{14} - 7 \times 10^{13}\, cm^{-3}$$

$$n_o - p_o = 3 \times 10^{13}\, cm^{-3} \qquad (1)$$

$$p_o = \frac{n_i^2}{n_o} = \frac{\left(2.5 \times 10^{13}\right)^2}{n_o} \qquad (2)$$

Substituting (2) into (1):

$$n_o - \frac{\left(2.5 \times 10^{13}\right)^2}{n_o} = 3 \times 10^{13}\, cm^{-3}$$

Multiply through by n_o and rearrange:

$$n_o^2 - 3 \times 10^{13} n_o - \left(2.5 \times 10^{13}\right)^2 = 0$$

Solution to quadratic equation:

$$n_o = \frac{3 \times 10^{13} \pm \sqrt{\left(3 \times 10^{13}\right)^2 + 4\left(2.5 \times 10^{13}\right)^2}}{2} = \frac{3 \times 10^{13} \pm 5.8 \times 10^{13}}{2} = 4.4 \times 10^{13} \quad \square \text{ disgard negative}$$

$$p_o = 1.4 \times 10^{13} \quad \square \text{ use either expression to calculate}$$

Charge Neutrality Equation

$$p_o + N_d^+ = n_o + N_a^-$$

Different from n_0–p_0

→ When close to the intrinsic concentration, we cannot neglect these carriers!

From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.
ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

Drift of Carriers in Electric Fields

Conductivity and Mobility

Drift Versus Random Motion

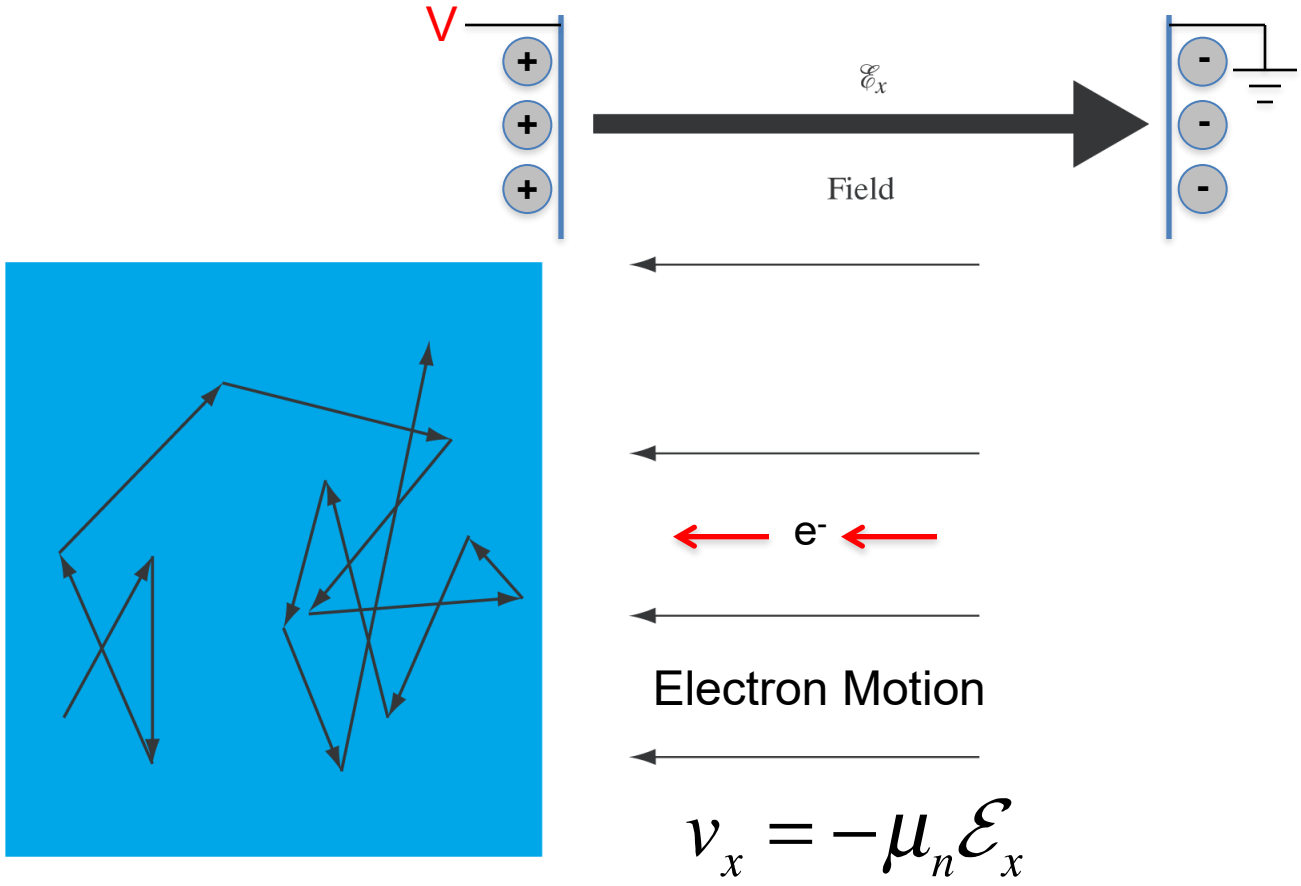


Figure 3.20

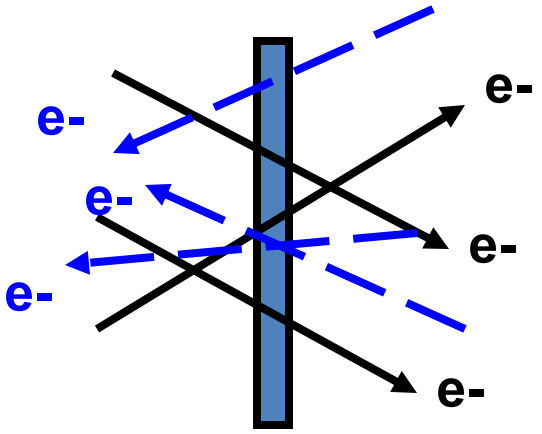
(a) Random thermal motion of an electron in a solid;
(b) well directed drift velocity with an applied electric field.

From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.
ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

Diffusion: A Brief Comment

- Diffusion: Net movement from higher concentration region to lower concentration region
- In the case of a uniform distribution of carriers, the random motion is averaged out (no net movement)
- If there is no net movement of carriers, there is no net current
- We will consider carrier drift under the condition where the distribution of carriers is uniform such that there is no diffusion contribution to the net current
  - We will come back to this assumption later

For Uniform e- Distribution



$$\text{Net "e}^-\text{" flow} = 0$$

$$\sum_i e_i^- \bullet v_i = 0$$

Drift in a Semiconductor

Assume uniform distribution of carriers: no diffusion

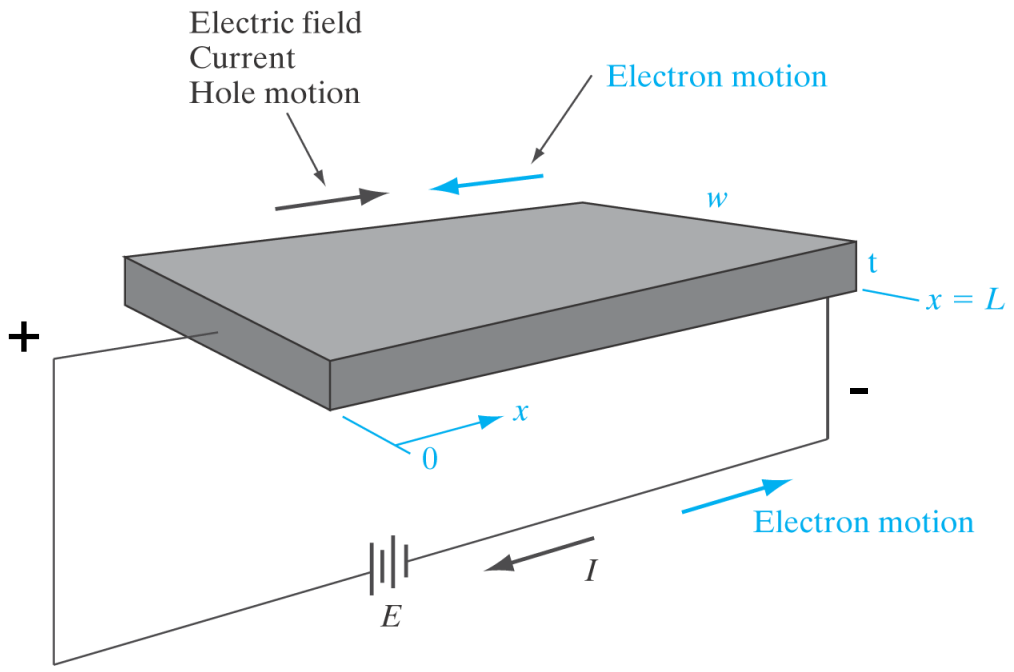


Figure 3.21

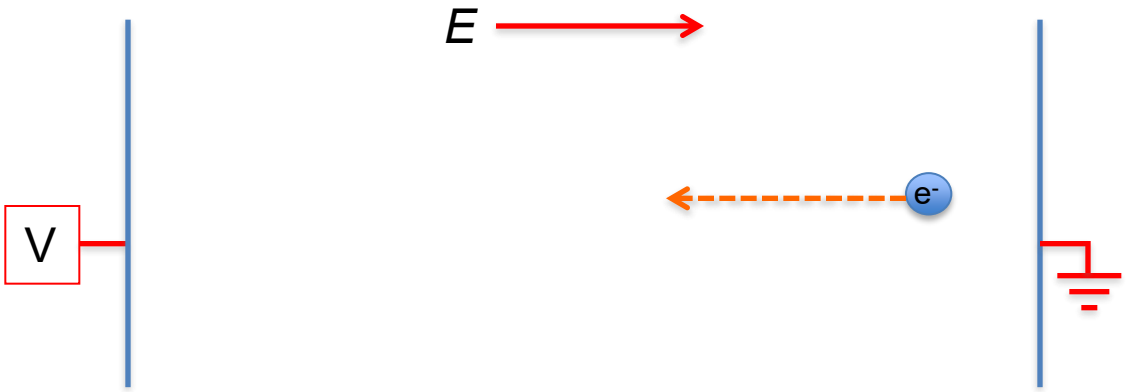
Drift of electrons and holes in a semiconductor bar.

From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.
ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

Electron Motion in a Crystal

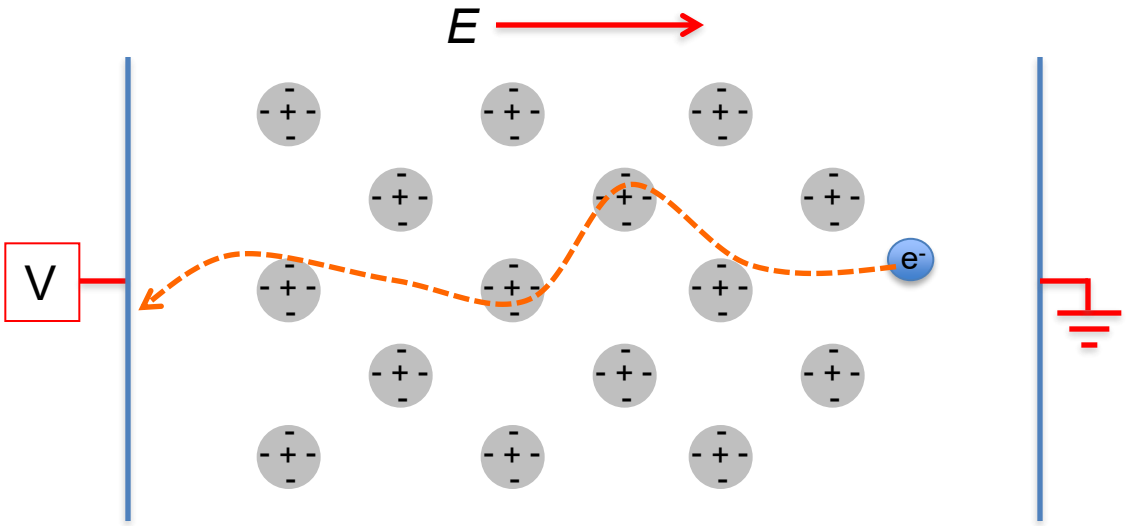
Case 1: in vacuum

- An electron in a vacuum will continue to accelerate under the influence of an electric field



Case 2: In crystal lattice

- An electron traveling through the crystal will experience the periodic potential of the lattice atoms
- The electron will be 'scattered'
- This will impart a momentum change to the electron
- A faster electron will scatter more quickly



Electron Drift in a Constant Electric Field

Force Due to Electric Field:

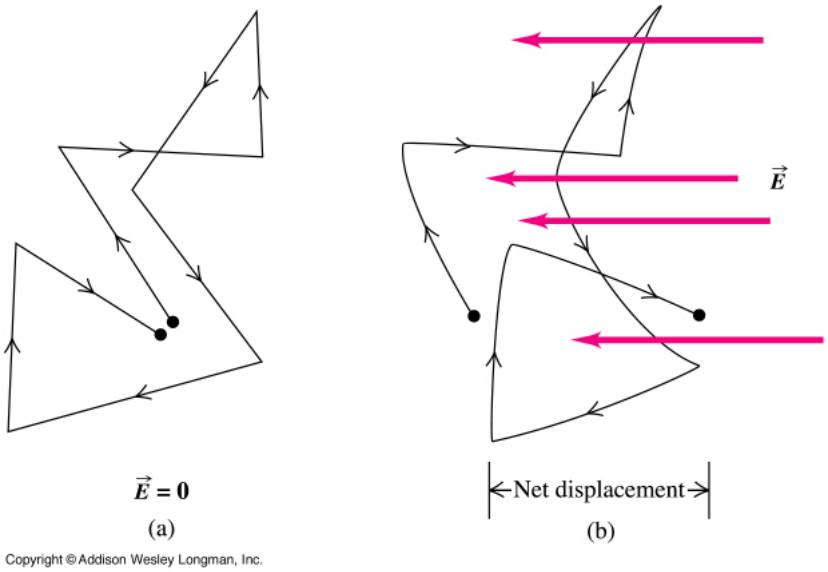
- Electric field creates a net force on each electron: **F = -qE**

$$\text{Force for "n" electrons} = (-q)nE_x = ma = m\left.\frac{dv_x}{dt}\right|_{field} = \left.\frac{dp_x}{dt}\right|_{field}$$

- This force creates a **net momentum** for the population of electrons in the direction of the field

There Must Be an Offsetting Force:

- Scattering events (electron collisions) limit the velocity of the electron
- The net rate of change of momentum, including collisions, is zero under steady-state conditions (otherwise, current would not be constant).



Copyright © Addison Wesley Longman, Inc.

Force for a single electron:

$$F = -qE_x$$

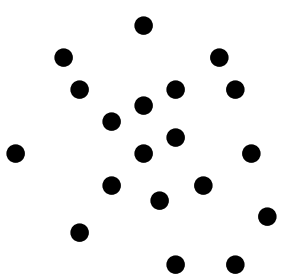
Force for "n" electrons:

$$-nqE_x = \left.\frac{dp_x}{dt}\right|_{field}$$

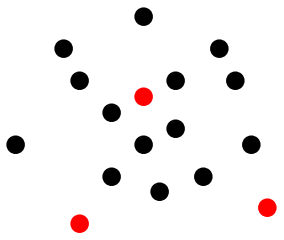
Mean Free Time and Collision Probability

- We start with an initial population of N_0 electrons (or holes) at time t=0
- We assume that any given electron has a fixed probability in time P of having a collision (scattering)
- If we think of scattering centers as fixed points, an electron with average velocity <v> travels an average distance <l> before colliding
- The average time between collisions is thus:

$$\langle t \rangle = \frac{\langle l \rangle}{\langle v \rangle} \equiv \bar{t}$$



Some time later "t"



N(t): # of electrons not collided.

- The probability in time Δt of having a collision is therefore given by:

$$P = \frac{\Delta t}{\bar{t}}$$

• Experienced Collision

Change in Carrier Population

If the probability of a single collision in an interval Δt is given by:

$$P = \frac{\Delta t}{\overline{t}}$$

The incremental number of electrons colliding

in an interval Δt is given by:

$$N(t) - N(t + \Delta t) = PN(t) = \frac{\Delta t}{\overline{t}} N(t)$$

Rewriting this:

$$\frac{N(t) - N(t + \Delta t)}{\Delta t} = \frac{1}{\overline{t}} N(t)$$

Taking the limit as $\Delta t \to 0$:

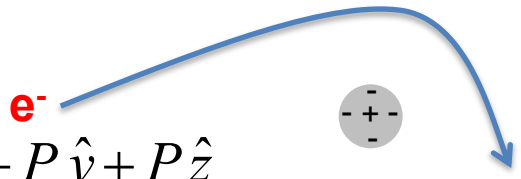
$$\lim_{\Delta t \to 0} \left(\frac{N(t) - N(t + \Delta t)}{\Delta t} \right) = - \lim_{\Delta t \to 0} \left(\frac{N(t + \Delta t) - N(t)}{\Delta t} \right) = - \frac{dN(t)}{dt} = \frac{1}{\overline{t}} N(t)$$

Momentum Transfer from Collisions

From the Previous Slide:

$$\frac{dN(t)}{dt} = -\frac{1}{\bar{t}}N(t) \text{ so:}$$

$$N(t) = N_0 e^{-t/\bar{t}}$$



$$\vec{P} = P_x\hat{x} + P_y\hat{y} + P_z\hat{z}$$

$$\vec{P} = \left(P_x - \langle p_x \rangle\right)\hat{x} + P_y\hat{y} + P_z\hat{z}$$

If the momentum change (loss) due to collisions for the entire population is $-p_x$ the incremental momentum change dp_x in a time interval dt is given by:

$$\left.\frac{dp_x}{dt}\right|_{collisions} = -\frac{p_x}{\bar{t}}$$

p_x: total momentum change if all electrons have collided (and randomized the directed momentum)

Note that the average momentum change per electron is given by: $\langle p_x \rangle = \frac{p_x}{n}$